SRM

SET 1

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

## Internal Assessment – III

Program: B.Tech
Course Code & Title: 18CYB101J/Chemistry
Year & Sem: I/II

Date: 20-06-2022
Time: 8.00-9.40am
Max. Marks: 50 Marks

### Part-A (10X1=10 Marks)

**Answer ALL the questions**

1. Chiral molecules are those which
   a) shows geometrical isomerism b) are superimposable on their mirror images c) are not superimposable on their mirror images d) are unstable molecules
2. Which of the following compounds will exhibit cis-trans isomerism?
   a)2-butanol b) 2-butene c) tartaric acid d) lactic acid
3. According to the convention, the Daniel cell is represented as _____
   a) Zn l $ZnSO_4$ ll $CuSO_4$ l Cu, E = 1.09 volt b) Zn l $ZnSO_4$ ll Cu l $CuSO_4$ , E = 1.09 volt c) $ZnSO_4$ l Zn ll $CuSO_4$ l Cu, E = 1.09 volt d) Zn l ZnS ll $CuSO_4$ l Cu, E = 1.09 volt
4. The entropy of an isolated system always ____ and reaches ____ when equilibrium is reached.
   a) remains constant, maximum b) decreases, minimum c) increases, maximum d) remains constant, minimum
5. In Pourbaix diagram, the redox reaction, $Fe^{2+} + 2e^- \rightarrow Fe_{(s)}$ is
   a) pH dependent b) pH independent c) solvent dependent d) potential independent
6. How many optical isomers are possible in a compound with one chiral carbon?
   a) 5 b) 4 c) 2 d) 3
7. Which of the following is not optically active compound?
   a) sugar solution b) Lactic acid c) Meso-tartaric acid d) Glyceraldehyde
8. The infinity of intermediate conformations is called
   a) Skew conformations b) Staggered conformations c) Eclipsed conformations d) Gauche
9. When the nucleophile $RO^-$ attacks the RX, the resultant product will be
   a) R – OH b) ROR c) R:CN d) RNHR
10. Identify the compound with the highest ring strain
    a) Cyclohexane b) Cyclopropane c) Cyclobutane d) Cyclopentane

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## Part- B (4X10 = 40Marks)

**Answer ANY FOUR questions**

11. (a) Give the differences between hard and soft acids. (4 marks)

    (b) Explain the mechanism of corrosion due to oxygen gas in dry atmosphere. (6 marks)

12. (a) Explain the following with examples (8 marks)

    i) Plane of symmetry ii) Centre of symmetry

    (b) Define solubility product ($K_{sp}$). (2 marks)

13. (a) Write a note on Geometrical isomerism. (5 Marks

    (b) Explain E2 mechanism taking an example. (5 Marks)

14. Write notes on oxidation and reduction reactions. (10 marks)

15. (a) Give the synthesis and uses of aspirin. (4 marks)

    (b) Write notes on hydrate and linkage isomerism. (6 marks)

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Department Of Chemistry
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Internal Assessment – III Answer Key

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wer ALL the MCQs Part-A (10*1=10 Marks)

1. c)
2. b)
3. a)
4. c)
5. b)
6. d)
7. c)
8. a)
9. b)
10. b)

t- B Answer ANY FOUR questions (4x10=40Marks)

11. (a) Give the differences between hard and soft acids.
**Any two differences with examples.** **(4marks)**

(b) Explain the mechanism of corrosion due to oxygen gas in dry atmosphere.
**Explanation with mechanism. [Main points are given]** **(6 marks)**

## Oxidation corrosion

- Occurs
  - due to direct chemical reaction of atm. $O_2$ with metal surface forming metal oxide
  - Absence of moisture
  - Increases with increase in temp.

Mechanism

- on exposure to atm., metal gets oxidized to form metal ions

(i) $M(S) \rightarrow M^{+n} + ne^-$

- Electrons lost by metal are taken up by oxygen to forms oxide ions

(ii) $1/2O_2(g) + 2e^- \rightarrow O^{2-}$

$$2M + n/2O_2 \longrightarrow 2M^{+n} + nO^{2-} \longrightarrow M_2O_n$$

Metal Oxide

12. (a) Explain the following with examples i) Plane of symmetry ii) Centre of symmetry
**Plane of symmetry for a molecule is the imaginary division, through which the molecule is bilaterally divided into two equal halves. These halves are mirror images of one another. Center of symmetry of a molecule is the center point from which the identical atoms exist on the opposite side from this center at equal distance. (2 marks)**
**Explanation for both with two examples for each. (2x3=6 marks)**

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(b) Define solubility product ($K_{sp}$).

**The solubility product constant is the equilibrium constant for the dissolution of a solid substance into an aqueous solution.** **(2 marks)**

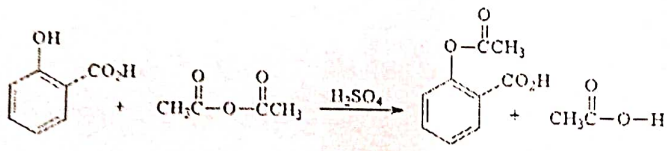
13. (a) Write a note on Geometrical isomerism.
**Explanation on geometrical isomerism with examples.** **(5 marks)**

(b) Explain E2 mechanism taking an example.
**It refers to bimolecular elimination and is basically a one-step mechanism.**
**Explanation of the mechanism with an example.** **(5 marks)**

14. Write notes on oxidation and reduction reactions.
**Taking the following reagents and explain its role by giving an example for oxidation and reduction reactions. [for each, one example with suitable equations]**
**(4x2.5=10 marks)**

**$KMnO_4$, $K_2Cr_2O_7$, $LiAlH_4$ and $NaBH_4$**

15. (a) Give the synthesis and uses of aspirin.



**(2 marks - No mechanism)**

**Any two uses for each.** **(2 marks)**

(b) Write notes on hydrate and linkage isomerism.

**Hydrate isomerism: They are also called as solvent isomerism and they differ in the number of water molecules but they have the same composition. In this one ion is directly attached to central metal but remaining is not. And they are called hydrated because when we talk about water, we use the word hydrated.**
**Linkage isomerism: Linkage isomers are defined as the same composition but differing in their connectivity to metals i.e same molecular formula but their attachment to metals is different.**
**Two examples for each.**
**(2x3=6 marks)**

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SET 2

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
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## Internal Assessment – III

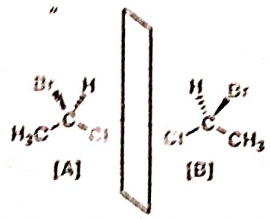
Program: B.Tech
Course Code & Title: 18CYB101J/Chemistry
Year & Sem: I/II

Date: 20-06-2022
Time: 8.00-9.40am
Max. Marks: 50 Marks

### Part-A (10X1=10 Marks)

**Answer ALL the questions**

1. The anode of the galvanic cell has _____
   a) Positive polarity b) Negative polarity c) No polarity d) Neutral
2. An example for hard acid is---,
   a) $Na^+$ b) RSH c) $OH^-$ d) $RNH_2$
3. The Helmholtz function (A/F) is given by
   a) U-TS b) U+TS c) -U-TS d) -U+TS
4. Anhydrous inorganic liquid metal surface in absence of moisture undergoes _____
   a) Wet corrosion b) Dry corrosion c) Galvanic corrosion d) Pitting corrosion
5. A centre of symmetry is equivalent to _____ fold alternating axis of symmetry.
   a) One b) Two c) Three d) Four
6. Which of the following compound would show optical isomerism?
   a) $H_2N\ CH(CH_3)_2$ b) $(CH_3)_2\ CHCHO$ c) $CH_3 - CH(OH)\ COOH$ d) $H_2N\ CH_2\ COOH$
7. The picture shown below is an example for --------




   a) Positional isomers b) mesomers c) geometric isomers d) enantiomers
8. The potential energy of n-butane is minimum for
   a) Skew conformations b) Gauche c) Eclipsed conformations d) Staggered conformations
9. Which step in $S_N1$ reaction is a slow rate determining step?
   a) Attack of nucleophile b) Formation of racemic mixture c) Formation of transition state
   d) both a and b
10. Aspirin is a ------- drug.
    a) Steroidal b) anti-depressant c) anti-biotic d) non-steroidal anti-inflammatory

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## Part- B (4X10 = 40Marks)

**Answer ANY FOUR questions**

11. (a) Define the terms i. Entropy ii. Enthalpy. (4 marks)

    (b) Write notes on free energy of a corrosion reaction. (6 marks)

12. (a) What is Galvanic cell? Give its representation. (4 marks)

    (b) Explain the type of isomerism exhibited by each of the following pairs: (6 marks)

    i) Maleic acid and Fumaric acid

    ii) n-Butyl alcohol and Diethyl ether

13. (a) What is Chirality? Differentiate Chirality and Achirality with an example for each. (5 marks)

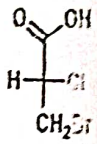
    (b) Write notes on ring opening reactions of cyclopropane. (5 marks)

14. (a) Compare addition reactions with substitution reactions. (5 marks)

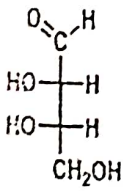
    (b) State the Markovnikov's rule and explain with an example. (5 Marks)

15. Explain the steps involved in Cahn Ingold Prelog rule in order to assign R/S notation and also apply it for all the chiral centers in the following molecules given in the form of Fischer projection. (10 Marks)

a)



b)



[top and bottom chiral centers]

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FN/ SET 2

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**Internal Assessment – III Answer Key**

Program: B.Tech
Course Code & Title: 18CYB101J/Chemistry
Year & Sem: I/II

Date: 20-06-2022
Time: 8.00-9.40am
Max. Marks: 50 Marks

**Answer ALL the MCQs Part-A (10*1=10 Marks)**

1. b)
2. a)
3. a)
4. b)
5. b)
6. c)
7. d)
8. d)
9. c)
10. d)

**Part- B Answer ANY FOUR questions (4x10=40Marks)**

11. (a) Define the terms i. Entropy ii. Enthalpy.
**Definitions for both.** **(2x2=4 marks)**

(b) Write notes on free energy of a corrosion reaction.

**The Gibb's free energy, given by the following equation, provides a tool with which to predict if a corrosion reaction is thermodynamically possible:**
**$\Delta G$ (Gibb's free energy in J) = - nFE**
**F= Faraday's constant (96,500 J/v – moles $e^-$); "n" = electrons transferred in oxidation reaction (mol $e^-$); E is electrode potential.**
**At standard conditions, $\Delta G^0 = - nFE^0$**
**If $\Delta G$ is positive, the reaction will not proceed. If $\Delta G$ is negative, the reaction is possible. Explain with an example.** **(6 marks)**

12. (a) What is Galvanic cell? Give its representation.

**An electrochemical cell that converts the chemical energy of spontaneous redox reactions into electrical energy is known as a galvanic cell or a voltaic cell.**

**$Zn \mid Zn^{2+}$ (1M) $\parallel Cu^{2+}$ (1M) $\mid Cu$** **(4 marks)**

**(Giving diagram is an option)**

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(b) Explain the type of isomerism exhibited by each of the following pairs: (2x3=6 marks)

i) **Maleic acid and Fumaric acid - Geometric isomers and explain on it.**

ii) **n-Butyl alcohol and Diethyl ether- functional isomerism and explain on it.**

13. (a) What is Chirality? Differentiate Chirality and Achirality with example.

**Definition for chirality.** **(2 marks)**

**Any one difference with an example for each.** **(3 marks)**

(b) Write notes on ring opening reactions of cyclopropane.

**Any two of the following:** **(2x2.5=5 marks)**

**Addition reactions with HI, Sulphuric acid, Halogens and Hydrogen**

14. (a) Compare addition reactions with substitution reactions.

**Comparison with suitable reactions as examples.** **(5 marks)**

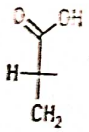
(b) State the Markovnikov's rule and explain with an example.

**When an unsymmetrical reagent is added to an unsymmetrical alkene, the negative part of the reagent gets attached to that carbon atom of the double bond which carries a lesser number of hydrogen atoms.**

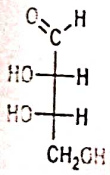
**Any one reaction with equation.** **(5 marks)**

15. Explain the steps involved in Cahn Ingold Prelog rule in order to assign R/S notation and also apply it for all the chiral centers in the following molecules given in the form of Fischer projection.

a)




b)




**Steps for CIP rule and give proper explanation. a) S b) 2S,3S** (top and bottom chiral centers) **(8+2=10 marks)**

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SET1

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## Internal Assessment – III

Program: B.Tech
Course Code & Title: 18CYB101J/Chemistry
Year & Sem: I/II

Date: 20-06-2022
Time: 10.00-11.40am
Max. Marks: 50 Marks

### Part-A (10X1=10 Marks)

**Answer ALL the questions**

1. Which of the following is not an example of chiral object?
   a) Cylindrical helix b) Square box c) Sandal or shoe d) Glove
2. Select the incorrect statement from the following option.
   a) The physical properties of enantiomers are identical
   b) In symmetrical environment, the chemical properties of enantiomers are identical
   c) The enantiomers react at same rate and form products in same amounts in asymmetrical environment d) Enantiomers have different solubilities in same chiral solvent
3. The plane which divides the molecule into two equal parts so that each half is the mirror image of other half is called
   a) Centre of symmetry b) Plane of symmetry c) Axis of symmetry d) Angle of symmetry
4. Decrease in free energy can be given by $-\Delta G =$ _____
   a) nFE b) n/FE c) nF/E d) F/nE
5. In Pourbaix diagram of iron one of the reactions depicted below is both potential and pH dependent
   a) $2Fe^{2+}+3H_2O \longrightarrow Fe_2O_3(s)+6H^++2e^-$ b) $Fe^{2+}+2e^- \longrightarrow Fe(s)$ c) $2Fe^{3+}+3H_2O \longrightarrow Fe_2O_3(s)+6H^+$
   d) $Fe^{3+}+e^- \longrightarrow Fe^{2+}$
6. The number of configurational isomers of molecules having (n) different chiral carbons is
   a) 2n b) $2^n$ c) $2^{n-1}$ d) $2^{n+1}$
7. Find the number of stereoisomers for $CH_3 - CHOH - CH = CH - CH_3$
   a) 1 b) 2 c) 3 d) 4
8. The potential energy of n-butane is maximum for
   a) Skew conformations b) Staggered conformations c) Eclipsed conformations
   d) Gauche
9. Which of the following reactions are favoured by polar aprotic solvent?
   a) $S_N1$ reactions b) $S_N2$ reactions c) Both $S_N1$ and $S_N2$ reactions d) Free radical reactions
10. $[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$ is an example for
    a) Coordination isomerism (b) Ionisation isomerism (c) hydrate isomerism
    (d) linkage isomerism

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## Part- B (4X10 = 40Marks)

**Answer ANY FOUR questions**

11. (a) What are Enantiomers and Diastereomers? Give examples. (4 marks)

(b) Explain the mechanism of hydrogen evolution type of corrosion. (6 marks)

12. (a) Derive Nernst equation and give any one application. (7 marks)

(b) What is Internal energy? Give its relation with enthalpy. (3 marks)

13. (a) Define alternating axis of symmetry with an example. (4 marks)

(b) Explain SN1 mechanism taking an example. (6 marks)

14. (a) Give the role of the following reagents in reduction and oxidation reactions.

i.$NaBH_4$ ii. $KMnO_4$ (6Marks)

(b) Give the synthesis and uses of Paracetamol. (4 marks)

15. Explain in detail the conformational analysis of n-butane with potential energy diagram. (10 Marks)

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AN / SET1

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**Internal Assessment – III Answer Key**

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**Year & Sem: I/II**

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**Time: 10.00-11.40am**
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**Answer ALL the MCQs Part-A (10*1=10 Marks)**

1. b)
2. c)
3. b)
4. a)
5. a)
6. b)
7. d)
8. c)
9. b)
10. a)

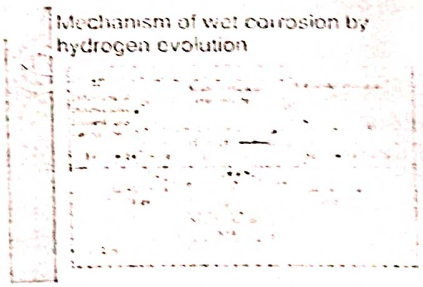
**Part- B Answer ANY FOUR questions (4x10=40Marks)**

12. (a) What are Enantiomers and Diastereomers? Give examples.

**Enantiomers are chiral molecules that are mirror images of one another and are not superimposable. Diastereomers are the stereoisomer compounds with molecules that are not mirrored images of one another and are not superimposable.** **[2 marks]**
**Any two examples for each.** **[2 marks]**

(b) Explain the mechanism of hydrogen evolution type of corrosion.

**Hydrogen evolution type of corrosion is nothing but displacement of H+ from acidic solution by metal ions. All the metals above $H_2$ in the electro chemical series get dissolved in acidic solution with simultaneous evolution of hydrogen.**
**At anode $Fe \rightarrow Fe^{2+} + 2e^-$ (oxidation /corrosion)**
**At cathode $2H^+ + 2e^- \rightarrow H_2$ (reduction)**
**Overall reaction $Fe + 2H^+ \rightarrow Fe$**
**Explanation with diagram.** **(6marks)**




13. (a) Derive Nernst equation and give any one application.

**The Nernst equation provides a relation between the cell potential of an electrochemical cell, the standard cell potential, temperature, and the reaction quotient.**
**Derivation** **(5 marks)**
**One application.** **(2marks)**

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(b) What is Internal energy? Give its relation with enthalpy

**Internal energy is the sum of all kinds of energies in a system. It is the total energy contained in a system. Enthalpy is the total heat content of a system.**

**Enthalpy is equal to the sum of internal energy and product of pressure-volume work.**

**$H = U + PV$**

**The first law of thermodynamics states that a change in internal energy is equal to the heat added to a system minus work done by the system.**

**$U = Q - W$**

**Both are state functions and both are extensive properties.** **(3 mark)**

14. (a) Define alternating axis of symmetry with an example.

**An improper or an alternating axis of symmetry of order n is an imaginary axis such that if rotation of the molecule around the axis by $360^0/n$ followed by reflection on a plane perpendicular to this axis generates a structure indistinguishable from the original one. An alternating axis of symmetry is designated as Sn. 'n' which is known as fold or order of the alternating axis is determined as follows:** **(3 marks)**

$$n = \frac{360^\circ}{\text{Angle of the rotation which satisfy the above condition}}$$

**Any one example.** **(1 mark)**

(b) Explain $S_N1$ mechanism taking an example.

**$S_N1$ reaction mechanism follows a step-by-step process wherein first, the carbocation is formed from the removal of the leaving group. Then the carbocation is attacked by the nucleophile. Finally, the deprotonation of the protonated nucleophile takes place to give the required product. The rate-determining step of this reaction depends purely on the electrophilicity of the leaving group and is not impacted at all by the nucleophile. (1 mark)**

**Mechanism taking an example.** **(5 marks)**

15. (a) Give the role of the following reagents in oxidation and reduction reactions.

i.$NaBH_4$ ii. $KMnO_4$

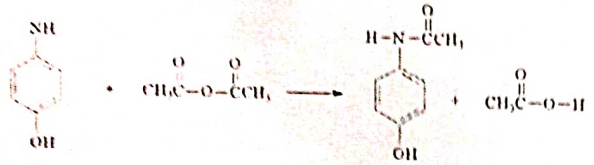
**i. and ii. are one of the examples for reducing and oxidising agents.** **(2 marks)**

**Give an example for both involved in reducing and oxidising reactions.**

**(With Equations 4 marks)**

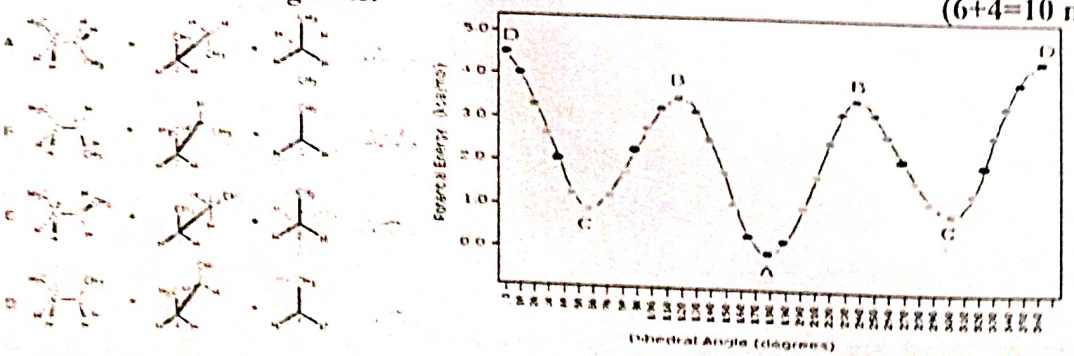
(b) Give the synthesis and uses of Paracetamol.

**Synthesis only (no mechanism) and any two uses.** **(2+2=4 marks)**



15. Explain in detail the conformational analysis of n-butane with potential energy diagram

**Explanation with diagrams.** **(6+4=10 marks)**

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SET 2

# DEPARTMENT OF CHEMISTRY

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## Internal Assessment – III

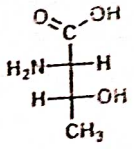
Program: B.Tech
Course Code & Title: 18CYB101J/Chemistry
Year & Sem: I/II

Date: 20-06-2022
Time: 10.00-11.40am
Max. Marks: 50 Marks

### Part-A (10X1=10 Marks)

Answer ALL the questions

1. Chiral molecules which are non-super-imposable mirror images of each other are called
   a) Diastereomers b) Meso compounds c) metamers d) Enantiomers
2. When a molecule has a plane of symmetry, it will be _____
   a) Optically inactive b) Optically active c) Both optically active and optically inactive d) Enantiomer
3. Generally, electrode potential refers to _____
   a) Reduction potential b) Oxidation potential c) Electron potential d) Cannot be determined
4. In an open system, for maximum work the process must be entirely
   a) irreversible b) reversible c) adiabatic d) both a and c
5. The following are state functions EXCEPT
   a) H b) q c) E d) S
6. The number of racemic forms of molecules having (n) different chiral carbons is
   a) 2n b) $2^n$ c) $2^{n-1}$ d) $2^{n+1}$
7. In gauche conformations, the methyl groups in n-butane are
   a) $60^0$ apart b) $90^0$ apart c) $180^0$ apart d) $360^0$ apart
8. The rate of nucleophilic substitution reactions is higher in the presence of
   a) Electron withdrawing groups b) Electron releasing groups
   c) both electron withdrawing and releasing groups d) radicals
9. Drugs that are used to diagnose, cure and prevent disease are called
   a) Pharmaceutical drugs b) addictive drugs c) industrial drugs
   d) single cell drugs
10. Assign the notation for the top and bottom chiral centres of the following molecule that is shown in the form of Fischer projection:




a) 1R, 3S b) 2S, 3S c) 2R, 3R d) 2S, 3R

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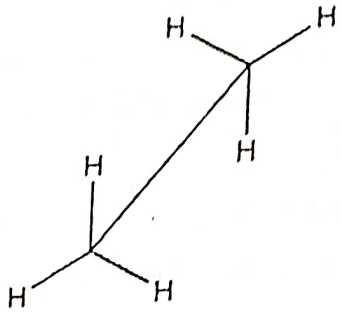
## Part- B (4X10 = 40Marks)

**Answer ANY FOUR questions**

11. (a) What are hard and soft bases? Give examples.
(b) Find the projection which is shown below and write a note on it.




12. Derive Gibbs-Helmoltz equation and give any one application. (10 Marks)

13. (a) Define standard electrode potential. Write the Nernst equation for the following cell reaction $Zn(s)/Zn^{2+}$ (aq) // $Cu^{2+}$ (aq) / Cu(s) (2 marks)

(b) Give the addition reaction products formed when Cyclopropane reacts with the following: (8 marks)

i) Halogens ii) HI iii) Sulphuric acid iv) Hydrogen

14. Discuss on types of isomerism exhibited in transition metal compounds with suitable examples. (10 Marks)

15.

16. (a) Explain free radical mechanism taking an example. (5 Marks)

(b) Explain Dieckmann condensation with an example. (5 Marks)

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AN/ SET 2

SRM

SRM Institute of Science and Technology
College of Engineering and Technology
Department Of Chemistry
SRM Nagar, Kattankulathur – 603203

**Internal Assessment – III Answer Key**

**Program: B.Tech** | **Date: 20-06-2022**
**Course Code & Title: 18CYB101J/Chemistry** | **Time: 10.00-11.40am**
**Year & Sem: I/II** | **Max. Marks: 50 Marks**

**Answer ALL the MCQs Part-A (10*1=10 Marks)**

1. d)
2. c)
3. a)
4. a)
5. a)
6. a)
7. b)
8. a)
9. b)
10. d)

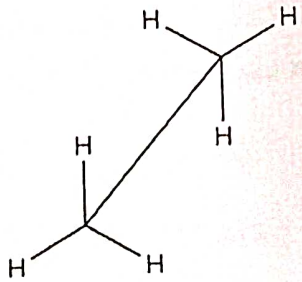
**Part- B Answer ANY FOUR questions (4x10=40Marks)**

12. (a) What are hard and soft bases? Give examples.

**Hard bases contain small, relatively nonpolarizable donor atoms and soft bases contain larger, relatively polarizable donor atoms.**

**Any two examples for each.** **[3+2=5 marks)**

(b) Find the projection which is shown below and write a note on it.




**Sawhorse projection** **(1 mark)**
**Explanation taking an example.** **(4 marks)**

13. Derive Gibbs-Helmoltz equation and give any one application.

**Derivation with one application.** **(8+2=10)'**

14. (a) Define standard electrode potential. Write the Nernst equation for the following cell reaction
$Zn(s)/Zn^{2+}(aq) // Cu^{2+}(aq) / Cu(s)$

**When the concentrations of all the species involved in a half-cell is unity (concentration is 1M) and temperature is 298 K, the potential difference develops between the electrode and the electrolyte (electrode potential) is known as standard electrode potential.**

**The given cell is Daniel cell consists of zinc and copper electrodes. The electrode reactions are:**

**Anode:** $Zn(s) \rightarrow Zn^{2+} + 2e^-$

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Cathode: $Cu^{2+} + 2e^- \rightarrow Cu(s)$

Net: $Zn(s) + Cu^{2+} \rightarrow Zn^{2+} + Cu(s)$

$E_{cell} = E_{cell}^0 - 2.303RT/2F \log [Zn(s)][Cu^{2+}] / [Cu(s)][Zn^{2+}]$

Since, the activities of pure copper and zinc metals are taken as unity, hence the Nernst equation for the Daniell's cell is,

$E_{cell} = E_{cell}^0 - 2.303RT/2F \log [Cu^{2+}] / [Zn^{2+}]$ (2 marks)

(b) Give the addition reaction products formed when Cyclopropane reacts with the following:
i. Halogens (1,3 dibromo propane and 1,3-dichlorpropane) ii. HI(n-propyliodide) iii. sulphuric acid (n-propanol) iv. Hydrogen (n-propane). Give equations. (4x2=8 marks)

14. Discuss on types of isomerism exhibited in transition metal compounds with suitable examples.
Any five of the following with two examples for each. (5x2=10 marks)
Hydrate, coordination, linkage, ionization, geometrical and optical isomerism.

16.(a) Explain free radical mechanism taking an example.
Free radical mechanism with steps taking an example. (5 marks)

(b) Explain Dieckmann condensation with an example.
The base-catalyzed intramolecular condensation of a diester to produce 5- or 6-membered cyclic ß- keto esters and is usually affected with sodium alkoxide in alcoholic solvent.
Mechanism taking an example. (5 marks)

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Kattankulathur – 603203

SET – 1

INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 10/12/2022
Duration: 8:00 – 09:00 AM
Max. Marks: 30

Part – A (10 x 1 = 10 Marks)
Answer ALL The Questions

1. Which of the following is a thermosetting polymer?
   a) polystyrene
   b) polyolefins
   c) nylons
   d) phenolic resins
2. The characteristics of condensation polymerization are given below
   I. only –C–C– linkages present in the polymer structure
   II. use of bifunctional or polyfunctional monomers
   III. elimination of a small byproduct molecule
   Which of the following is true?
   a) I, II, III
   b) II and III
   c) I and II
   d) Only III
3. Which of the following is true for the resultant polymer product formed, when molecules of phthalic acid react with molecules of glycerol?
   a) branch polymer
   b) cross-link polymer
   c) linear polymer
   d) none of the mentioned
4. Which of the following category does cellulose nitrate fall into?
   a) natural
   b) synthetic
   c) semi-synthetic
   d) none of the mentioned
5. Polymers are not classified on the basis of which of the following?
   a) Source
   b) Number of monomers
   c) Method of preparation
   d) Structure

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6. Which of the following is not a natural polymer?
   a) Rayon
   b) Starch
   c) Cellulose
   d) RNA
7. The compound $[-CH_2-CH(C_6H_5)^-]_n$ is a _____
   a) homopolymer
   b) co-polymer
   c) condensation polymer
   d) network polymer
8. _____ is the property of recovering original shape after the removal of def strain.
   a) Rigidity modulus
   b) Youngs modulus
   c) Elasticity
   d) Bulk modulus
9. Minimum interplanar spacing required for Bragg's diffraction is _____
   a) $\lambda/4$ b) $\lambda/2$ c) $4\lambda$ d) $2\lambda$
10. In XPS, the primary and secondary beams consist of
   a) X-ray photon, electron
   b) electrons, X-ray photon
   c) electrons, electrons
   d) UV-photons, electrons

## Part – B (2 x 10 = 20 Marks)

11. a. Discuss the addition polymerization process in the synthesis of PVC and polystyrene along with the properties and applications. (10 Ma

(OR)

b. i. Explain polymer tacticity and its types.
   ii. Write a short note on the synthesis and properties of synthetic rubber. (6 Mar

12. a. i. Differentiate thermoplastic and thermosets. (4 Mar
   ii. Write a short note on synthesis and properties of a conducting polymer. (6 Ma

b. Discuss the principle, instrumentation, and applications of XPS. (4 Ma

(OR) (10 M

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**DEPARTMENT OF CHEMISTRY**
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**SET – I**

**INTERNAL ASSESSMENT – III**

**Program: B.Tech**
**Course Code & Title:21CYB101J & Chemistry**
**Year & Sem: I Year & I Sem**

**Date: 10/12/2022**
**Duration: 8:00 – 09:00 AM**
**Max. Marks: 30**

**Part – A (10 x 1 = 10 Marks)**
**Answer ALL The Questions**

1. Which of the following is a thermosetting polymer?
   d) phenolic resins
2. The characteristics of condensation polymerization are given below-
   I. only −C−C− linkages present in the polymer structure
   II. use of bifunctional or polyfunctional monomers
   III. elimination of a small byproduct molecule
   Which of the following is true?
   b) II and III
3. Which of the following is true for the resultant polymer product formed, when molecules of phthalic acid react with molecules of glycerol?
   b) cross-link polymer
4. Which of the following category does cellulose nitrate fall into?
   c) semi-synthetic
5. Polymers are not classified on the basis of which of the following?
   b) Number of monomers
6. Which of the following is not a natural polymer?
   a) Rayon
7. The compound $[-CH_2-CH(C_6H_5)-]_n$ is a _____
   a) homopolymer
8. _____ is the property of recovering original shape after the removal of deforming strain.
   c) Elasticity
9. Minimum interplanar spacing required for Bragg's diffraction is _____
   b) $\lambda/2$
10. In XPS, the primary and secondary beams consist of
    **a) X-ray photon, electron**

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## Part – B (2 x 10 = 20 Marks)

11. a. Discuss the addition polymerization process in the synthesis of PVC and polystyrene along with the properties and applications. (10 Marks)

**Addition polymerization – Explanation (2 Marks)**
**Synthesis properties and applications of PVC (4 Marks)**
**Synthesis properties and applications of polystyrene (4 Marks)**

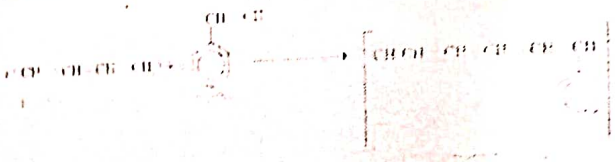
(OR)

b. i. Explain polymer tacticity and its types? (6 Marks)

**Types of tacticty: Isotactic, Syndiotactic and Atactic polymers (3 Marks)**
**Structures showing different tacticity types (3 Marks)**

ii. Write a short note on the synthesis and properties of synthetic rubber. (4 Marks)

**Synthesis: Reaction of butadiene with styrene giving rise to styrene-butadiene rubber. (1 Marks)**
**Reaction (1 Mark)**



**Properties (1 Marks)**

12. a. i. Differentiate thermoplastic and thermosets? (6 Marks)

**Differences between thermoplastic and thermosets with respect to structure, properties and molding ability or examples (6 Marks)**

ii. Write a short note on synthesis and properties of a conducting polymer. (4 Marks)

**Synthesis and properties of Polyacetylene or Poly(3-hexylthiophene) (4 Marks)**

(OR)

b. Discuss the principle, instrumentation, and applications of XPS (10 Marks)

**Principle of XPS (3 Marks)**
**Discussion related Instrumentation (3Marks)**
**Importance of Ultrahigh vacuum (1 Marks)**
**Diagram (3 Marks)**
**Applications (Optional)**

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SRM

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SET – II

INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 10/12/2022
Duration: 8:00 – 09:00 AM
Max. Marks: 30

Part – A (10 x 1 = 10 Marks)
Answer ALL The Questions

1. Which of the following polymer type is not classified on the basis of its application and properties?
   a) rubbers
   b) plastics
   c) fibres
   d) synthetic
2. Which of the following monomers are unsuitable for condensation polymerization?
   a) propanoic acid and ethanol
   b) butane-dioic acid and glycol
   c) diamines and dicarboxylic acids
   d) hydroxy acids
3. Which of the following kind of polymers are known for their high crystallinity?
   a) isotactic
   b) syndiotactic
   c) atactic
   d) gauche tactic
4. Which of the following is a co-polymer?
   a) Polythene
   b) Bakelite
   c) PVC
   d) Polyacrylonitrile
5. Which of the following types of polymers is not based on the classification by the source?
   a) Natural
   b) Semi-synthetic
   c) Elastomers
   d) Synthetic
6. The synthesis of which of the following polymers involves the repeated loss of small molecules?
   a) Polythene
   b) Buna-S

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c) Buna-N
d) Nylon-6,6

7. _____ undergoes permanent deformation on heating.
   a) Polythene
   b) PVC
   c) Teflon
   d) Bakelite

8. The strength of the polymer increases with _____ in molecular weight.
   a) Increase
   b) Decrease
   c) No change
   d) Slightly decrease

9. X-ray diffractometers are not used to identify the physical properties of which of the following?
   a) Metals b) Liquids c) Polymeric materials d) Solids

10. The source for XPS is -----
   a) Mercury - arc
   b) Nernst glower
   c) Globar source
   d) AlKα

## Part – B (2 x 10 = 20 Marks)

11. a. Discuss the condensation polymerization process in the synthesis of Nylon and Polyurethane along with their properties and applications. (10 Marks)

(OR)

b. i. How polymers are classified based on origin and nomenclature? (6 Marks)
   ii. Define degree of polymerization and functionality? (4 Marks)

12. a. i Discuss the synthesis, properties, and applications of polypropylene (4 Marks)
   ii. Explain n and p doping in conducting polymers. (6 Marks)

(OR)

b. i. Explain Bragg's law with a neat diagram (6 Marks)
   ii. Compute the Miller Indices for a plane intersecting at x= ¼, y=1, and z=1/2 (4 Marks)

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SET – II

INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 10/12/2022
Duration: 8:00 – 09:00 AM
Max. Marks: 30

Part – A (10 x 1 = 10 Marks)
Answer ALL The Questions

1. Which of the following polymer type is not classified on the basis of its application and properties?
   a) rubbers
   b) plastics
   c) fibres
   d) synthetic
2. Which of the following monomers are unsuitable for condensation polymerization?
   a) propanoic acid and ethanol
   b) butane-dioic acid and glycol
   c) diamines and dicarboxylic acids
   d) hydroxy acids
3. Which of the following kind of polymers are known for their high crystallinity?
   a) isotactic
   b) syndiotactic
   c) atactic
   d) gauche tactic
4. Which of the following is a co-polymer?
   a) Polythene
   b) Bakelite
   c) PVC
   d) Polyacrylonitrile
5. Which of the following types of polymers is not based on the classification by the source?
   a) Natural
   b) Semi-synthetic
   c) Elastomers
   d) Synthetic
6. The synthesis of which of the following polymers involves the repeated loss of small molecules?
   a) Polythene
   b) Buna-S

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c) Buna-N
d) Nylon-6,6

7. _____ undergoes permanent deformation on heating.
   a) Polythene
   b) PVC
   c) Teflon
   d) Bakelite

8. The strength of the polymer increases with _____ in molecular weight.
   a) Increase
   b) Decrease
   c) No change
   d) Slightly decrease

9. X-ray diffractometers are not used to identify the physical properties of which of th following?
   a) Metals b) Liquids c) Polymeric materials d) Solids

10. The source for XPS is -----
   a) Mercury - arc
   b) Nernst glower
   c) Globar source
   d) AlKα

## Part – B (2 x 10 = 20 Marks)

11. a. Discuss the condensation polymerization process in the synthesis of Nylon a Polyurethane along with their properties and applications. (10 Marks)

(OR)

b. i. How polymers are classified based on origin and nomenclature? (6 Marks)
   ii. Define degree of polymerization and functionality? (4 Marks)

12. a. i Discuss the synthesis, properties, and applications of polypropylene (4 Mar
   ii. Explain n and p doping in conducting polymers. (6 Mar

(OR)

b. i. Explain Bragg's law with a neat diagram (6 Mar
   ii. Compute the Miller Indices for a plane intersecting at x= ¼, y=1, and z=1/2 (4 Mar

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# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
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Kattankulathur – 603203

Set-I

## INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 10/12/2022
Duration: 12:30 – 1:30 PM
Max. Marks: 30

**Part – A (10 x 1 = 10 Marks)**
**Answer all the questions**

1 One of the characteristic properties of polymer material is
(a) High temperature stability
(b) High mechanical strength
(c) High elongation
(d) Low hardness

2 Find the incorrect macromolecule among the following
a) Cellulose
b) Glycogen
c) RNA
d) Galactose

3 Which of the following is a thermosetting polymer?
a) polystyrene
b) polyolefins
c) nylons
d) phenolic resins

4 Which of the following kind of polymers are known for their high crystallinity?
a) isotactic
b) syndiotactic
c) atactic
d) none of the mentioned

5 Glass transition temperature (Tg) for Nylon-6 is 50°C, which is higher than polyethylene due to _____
a) Inter-molecular hydrogen bonding
b) Intra-molecular hydrogen bonding
c) Vander Waals forces
d) Covalent forces

6 _____ is the property of recovering original shape after the removal of deforming strain.
a) Rigidity modulus
b) Young's modulus
c) Elasticity

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d) Bulk modulus

7 The neighboring polymeric chains in thermosets are held together by _____
a) Van der Waals force
b) Hydrogen bond
c) Covalent bond
d) Electrovalent bond

8 Which of the following is not an application of conducting polymers?
a) Rechargeable batteries
b) Analytical sensors
c) Electronics
d) Adhesives

9 The X-ray source for XPS is
a) Mercury- arc
b) Nernst glower
c) Globar source
d) Al Kα

10 Obtain the Miller indices of a plane whose intercepts are 4, 4 and 2 units along the three axes.
a) (122)
b) (211)
c) (121)
d) (112)

**Part – B (2 x 10 = 20 Marks)**

11. a. Discuss the condensation polymerization process in the synthesis of Nylon and PET along with their properties and applications. (10 Marks)

(OR)

b. i. What are the differences between thermoplastic and thermosets? Give examples. (8 Marks)

ii. Give an example of copolymer. (2 Marks)

12. a. i Discuss the synthesis, properties and applications of polypropylene. (4 Marks)

ii. Explain n- and p- doping in conducting polymers. (6 Marks)

(OR)

b. Discuss the principle and instrumentation of X-ray photo electro spectroscopy. (10 Marks)

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SRM INSTITUTE OF SCIENCE & TECHNOLOGY

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
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Kattankulathur – 603203

Set-II

## INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 10/12/2022
Duration: 12:30 – 1:30 PM
Max. Marks: 30

**Part – A (10 x 1 = 10 Marks)**

**Answer all the questions**

1 Gutta-percha is a
a) syndiotactic polymer
b) isotactic polymer
c) Atactic polymer
d) Eutactic polymers

2 Identify the polymer formed by addition polymerization
a) Nylon
b) Polyester
c) Teflon
d) Bakelite

3 The synthesis of which of the following polymers involves the repeated loss of small molecules?
a) Polythene
b) Buna-S
c) Buna-N
d) Nylon-6,6

4 _____ undergoes permanent deformation on heating.
a) Polythene
b) PVC
c) Teflon
d) Bakelite

5 If the polymer is in the room temperature, then it is _____
a) Brittle
b) Viscofluid state
c) Amorphous
d) Rubbery

6 Minimum interplanar spacing required for Bragg's diffraction is
a) $\lambda/4$ b) $\lambda/2$
c) $4\lambda$ d) $2\lambda$

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7 Select the incorrect statement from the following option.
a) Thermosets have 3-dimensional, cross-linked network structure
b) Thermosets cannot be remoulded, reused or reclaimed
c) Thermosets are hard, strong and brittle
d) Thermosets are soluble in suitable solvents

8 Which of the following is used for making rechargeable batteries?
a) Polypyrrole
b) Polyester
c) Polyaniline
d) Polyacrylonitrile

9 Aramid fibers are used for _____
a) Apparels
b) Shoe soles
c) Packing materials
d) Bullet proof jackets

10 Haemodialysis tubes are made with _____
a) Thermoplastic polyurethane
b) Polyurethane intermediate
c) Polystyrene
d) Silicone rubber

**Part – B (2 x 10 = 20 Marks)**

11. a. Discuss the addition polymerization process in the synthesis of PVC, Teflon along with their properties and applications. (10 Marks)

(OR)

b. i. Explain polymer tacticity and its types? (6 Marks)
ii. Write a short note on the synthesis and properties of synthetic rubber. (4 Marks)

12. a. i. Explain Bragg's law with a neat diagram. (6 Marks)
ii. For the intercepts x, y, and, z with values of 3, 1, and 2 respectively, find the Miller indices. (4 Marks)

(OR)

b. i How polymers are classified based on structure and nomenclature? (6 Marks)
ii. Write a short note on synthesis and properties of a conducting polymer. (4 Marks)

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# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
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SET – I

## INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title:21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 13/12/2022
Duration: 08.00 – 09.00 AM
Max. Marks: 30

*Part – A (10 x 1 = 10 Marks)*
*Answer ALL The Questions*

1. Which of the following monomers are unsuitable for condensation polymerization?
   a) propanoic acid and ethanol b) butane-dioic acid and glycol c) diamines and dicarboxylic acids d) hydroxy acids
2. Pthalic acid reacts with glycerol to give ……..
   a) branch polymer b) cross linked polymer c) linear polymer d) graft polymer
3. Buna-N rubber is ……..
   a) Styrene-butadiene b) Chloroprene c) Neoprene d) Acrylonitrile butadiene
4. Consider the following statements for condensation polymerization :
   I. Bifunctional or polyfunctional monomers
   II. Loss of each kind of functional group in each step for bifunctional species
   III. Always accompanied by the release of a byproduct molecule
   IV. Monofunctional or polyfunctional monomers
   Which of the following are true?
   a) I and IV b) I, II and III c) I and II d) III and IV
5. Which of the following monomers cannot undergo chain growth polymerization?
   a) $CH_2$= $CH_2$ b) $CH_2$=CHCN c) COOH–$CH_2$–COOH d) $CH_2$=CHCOOR
6. Which one of the below can be used as an insulator and also as a lubricant?
   a) Polypropylene b) PTFE c) Nylon d) Polyurethane
7. Intermolecular forces of thermoplastic polymers are ……
   a) more than elastomers b) between elastomers and fibres c) same as elastomers d) more than fibres
8. Glass transition temperature ($T_g$) for Nylon-66 is 50 °C, which is higher than polyethylene due to _____
   a) Vander Waals forces b) covalent bonding c) Inter-molecular hydrogen bonding d) Intra-molecular hydrogen bonding
9. Minimum interplanar spacing required for Bragg's diffraction is ……
   a) λ/4 b) λ/2 c) 2λ d) λ
10. Which among the following polymers have lowest solubility?
    a) polyethylene b) polystyrene c) epoxy resin d) nylon-66

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## *Part – B (2 x 10 = 20 Marks)*

11. a. Write the preparation, properties, and applications of the following polymers: (10 marks)
i. PTFE ii. Polyurethane and iii. Nylon-66

**(OR)**

b. i. Compare and contrast syndiotactic and isotactic polymers. Provide suitable examples. (6 marks)

ii. Elucidate with an example 'p-doping' conducting polymer (4 marks)

12. a. With a neat sketch, discuss the principle, instrumentation and applications of XPS. (10 marks)

**(OR)**

b. i. Define Miller indices. Compute the Miller indices for a plane intersecting at x= 1/4, y=1, and z=1/2. (4 marks)

ii. Suggest the products and provide suitable equations when 1, 3 – butadiene reacts with the following i. Acrylonitrile ii. Styrene (6 marks)

*********

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# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

SET – II

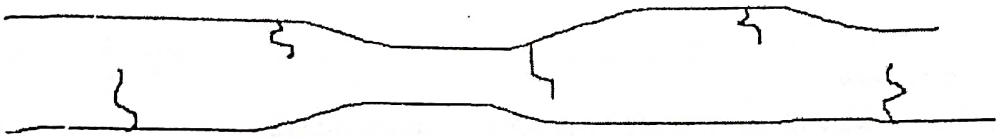
## INTERNAL ASSESSMENT – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 13/12/2022
Duration: 08.00 – 09.00 AM
Max. Marks: 30

*Part – A (10 x 1 = 10 Marks)*
*Answer ALL The Questions*

1. Which of the following polymers has a structure as shown below?



a) Low-density polythene b) High-density polythene c) Polyvinyl chloride d) Bakelite

2. Which of the following does not have dipole-dipole interactions?
   a) Nylon b) Terylene c) Dcaron d) Nylon
3. Buna-S rubber is .....
   a) Styrene-butadiene b) Chloroprene c) Neoprene d) Acrylonitrile butadiene
4. The reaction mixture of addition polymerization contain at any instant of time :
   I. Full grown polymer molecules
   II. Unreacted monomer molecules
   III. Free radical chains and initiators
   Which of the above statements are correct?
   a) I and II b) I, II and III c) I and III d) III only
5. Monomer used in the synthesis of neoprene is .....?
   a) $CH_2$= CHCl b) $CH_2$=CHCN c) $CH_2$= CCl-CH= $CH_2$ d) $CCl_2$=$CCl_2$
6. Hemodialysis tubes are made with .....
   a) Polypropylene b) Thermoplastic polyurethane c) Nylon d) Polyurethane intermediate
7. Elastomers have the general structure of .....
   a) flexible linear chains b) rigid three dimensional network c) linear cross linked chains d) rigid linear chains
8. Glass transition temperature ($T_g$) for Nylon-66 is 50 °C, which is higher than polyethylene due to _____
   a) Vander Waals forces b) covalent bonding c) Inter-molecular hydrogen bonding d) Intra-molecular hydrogen bonding

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9. If X-ray of wavelength 100 Å is incident on an atom at an angle of 90°, then what shou be the value of d for first-order spectrum?
   a. 30 Å  b. 40 Å  c. 50 Å  d. 60 Å
10. Which of the following kind of polymers are known for their high crystallinity?
   a) random  b) isotactic  c) atactic  d) syndiotactic

## *Part – B (2 x 10 = 20 Marks)*

11. a. Write the preparation, properties, and applications of the following polymers: (10 marks)
   i. Polypropylene  ii. Polystyrene  iii. PET

(OR)

   b. i. Compare and contrast addition and condensation polymerisation. Provide suitable examples (6 marks)
   ii. Elucidate with an example 'n-doping' conducting polymer (4 marks)

12. a. i. Explain the principle and applications of XPS. (6 marks)
   ii. Compute the Miller indices for a plane intersecting at x= ¼, y=1, and z=1/2.

(OR)

   b. i. With a neat sketch derive Bragg's law. (5 marks)
   ii. How polymers are classified based on origin and nomenclature? (5 marks)

*********

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DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
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Kattankulathur – 603203

SET 1

INTERNAL ASSESSMENT – III

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Semester: I Year & I Semester

Date: 13/12/2022
Duration: 12.30 – 1.30 pm
Max. Marks: 30

**Part- A (10 x 1 = 10 Marks)**

**Answer ALL the Questions**

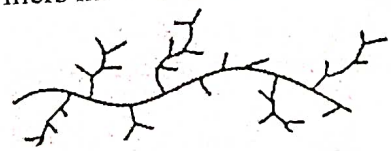
1. If the angle of incidence is 45°, then the wavelength for first-order spectral signal is equal to -------
   (a) d (b) 1.41d (c) 2d (d) 0.71 d

2. X-ray diffractometers are not used to identify the physical properties of which of the following?
   (a) Metals (b) Liquids (c) Polymeric Materials (d) Solids

3. The source of X-rays used in XPS is -----
   (a) $CuK_\alpha$ (b) Nernst glower (c) $MgK_\alpha$ (d) Mercury – arc

4. The kinetic energy of the photoelectron energies is dependent on _____ of the atom, which makes XPS useful to identify the oxide state.
   (a) Mass (b) Charge (c) Chemical environment (d) Volume

5. Which of the following polymers has a structure as shown?



   (a) Low density polyethylene
   (b) High density polyethylene
   (c) Polystyrene
   (d) Polypropylene

6. Which of the following is not an elastomer?
   (a) Buna-S (b) Buna-N
   (c) PVC (d) Neoprene

7. _____ undergoes permanent deformation on heating.
   (a) Polythene (b) PVC (c) Teflon (d) Bakelite



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8. Glass transition temperature (Tg) for Nylon-6 is 50°C, which is higher than polyethylene due to _____
(a) Inter-molecular hydrogen bonding
(b) Intra-molecular hydrogen bonding
(c) Vander Waals forces
(d) Covalent forces

9. Which of the below polymers show higher crystallinity ?
(a) Syndiotactic PP (b) Atactic PP (c) Isotactic PP (d) LDPE

10. Which of the following is not an application of conducting polymers?
(a) Rechargeable batteries (b) Analytical sensors
(c) Electronics (d) Adhesives

**Part- B (2 x 10 = 20 Marks)**

11 **a. i.** Discuss the synthesis, properties and applications of (i) PTFE and (ii) Nylon 66 (10 Marks)

**OR**

**b. i.** Draw a neat diagram showing the components of X-ray Photoelectron Spectrophotometer and explain its working. (10 Marks)

12 **a. i.** Define Miller indices with example. Draw and compute the Miller Indices for a plane intersecting at x= 1/3, y=2, and z=1/2. (6 Marks)

**ii.** Is $-[-CH_2-CH(C_6H_5)-]_n-$ a homopolymer or copolymer? Write the name and structure of monomer. Which method can be used to synthesize this polymer? (4 Marks)

**OR**

**b. i.** Write a note on conducting polymers. Give two examples with their structure. (6 Marks)

**ii.** Compare Natural and Synthetic rubbers. (4 Marks)

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SRM Institute of Science and Technology
Kattankulathur – 603203

SET 2

INTERNAL ASSESSMENT – III

**Program: B. Tech.**
**Course Code & Title: 21CYB101J & Chemistry**
**Year & Sem: I Year & I Sem**

**Date: 13/12/2022**
**Duration: 12.30 – 1.30 PM**
**Max. Marks: 30**

**Part- A (10 x 1 = 10 Marks)**

**Answer ALL the Questions**

1. If X-ray of wavelength 20 Å is incident on crystalline solid at an angle of 30°, then what should be the value of 'd' for first-order spectral signal?
(a) 20 Å (b) 10 Å (c) 40 Å (c) 30 Å

2. Repeatable entity of a crystal structure is known as
(a) Crystal (b) Lattice (c) Unit cell (d) Miller indices

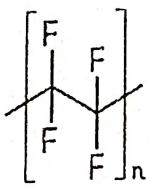
3. In XPS, the primary and secondary beams consist of
(a) X-ray photon, electron (b) electrons, X-ray photon
(c) electrons, electrons (d) UV-photons, electrons

4. The kinetic energy of the ejected photoelectron is dependent upon the energy of which of the following?
(a) Ions around (b) Photons around (c) Material (d) Impinging photon

5. The compound $[-CH_2-CH(C_6H_5)^-]_n$ is a _____
(a) homopolymer (b) co-polymer
(c) condensation polymer (d) network polymer

6. Which of the following has the weakest intermolecular forces?
(a) Buna-N (b) Nylon-6,6
(c) Polythene (d) Polystyrene

7. The polymer shown is a _____.




(a) elastomer (b) fibre (c) thermoplastic (d) thermosetting plastic



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8. Organic solvent containers are made with .....
(a) Polypropylene (b) PET (c) Polystyrene (d) PVC

9. Weak intermolecular attraction forces are seen in .....
(a) Thermosets (b) Thermoplastic (c) Elastomers (d) Fibers

10. What is the trade name for polyisoprene?
(a) Natural rubber (b) Neoprene (c) Silicone (d) Buna acrylonitrile

**Part- B (2 x 10 = 20 Marks)**

11 a. i. Discuss the synthesis, properties and applications of (i) Polystyrene and (ii) PET (10 Marks)

**OR**

b. i. Derive the expression for Bragg's law and calculate the minimum interplanar spacing required for Bragg's diffraction. Explain, why X-ray diffractometers are not used to identify the physical properties of the liquids. (10 Marks)

12 a. i. Discuss the principle of XPS with suitable diagram and explain why the technique is also called as ESCA. (6 Marks)

ii. Draw the structure of the monomer for each of the following polymers
(A) Poly(vinyl chloride) (B) Polyurethane (C) Nylon 66 (4 Marks)

**OR**

b. i. Explain n type and p type doping conducting polymers and give their applications. (6 Marks)

ii. Compare thermo and thermosetting polymers. (4 Marks)

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SET 1

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

CLA 3

Program: B.Tech
Course Code & Title: 21CYB101J / Chemistry
Year & Sem: I/II

Date: 04-05-2023
Time: 8.00-9.00am
Max. Marks: 30

**Answer ALL the MCQs Part-A (10 x 1=10 Marks)**

1. Which of the following is an initiator molecule in the free radical polymerisation?

a) Sulphuric acid b) Benzoyl peroxide c) Potassium permanganate d) Chromium oxide

2. The IUPAC name for paracetamol is

a) 2-Acetoxybenzoic acid b) Monohydroxybenzene c) N-(4- hydroxyphenyl) acetamide d) Phenyl Salicylate

3. Which of the following are thermoplastic?

i) Teflon ii) Bakelite iii) Vulcanised rubber iv) Polystyrene

a) ii only b) i and iv c) ii and iii d) i only

4. The chemical name of Nylon 6,6 is

a) Poly vinyl chloride b) Poly caprolactum c) cis-poly isoprene

d) Polyhexamethyleneadipamate

5. Natural rubber is basically a polymer of

a) Neoprene b) Isoprene c) Chloroprene d) Butadiene

6. In addition polymer, monomer used is

a) Saturated compounds b) Unsaturated compounds c) Bifunctional saturated compounds

d) Trifunctional saturated compounds

7. Usually the stronger constituent of a composite is

a) Matrix b) Coating c) Laminar d) Reinforcement

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8. Which of the following is an application of glass-fibre reinforced composites?

a) Adhesives b) Conveyor belts c) Design of ships d) Automotive parts

9. XPS focuses on which of the following information?

a) Mass of the electron b) Charge of the electron c) Binding energy of the electron

d) Mass of the atoms

10. Minimum interplanar spacing required for Bragg's diffraction is _____

a) $\lambda/4$ b) $\lambda/2$ c) $4\lambda$ d) $2\lambda$

**Part-B (2 x 10= 20 Marks)**

11. a) i.Explain the mechanism followed in the formation of anti-Markovnikov's product n-propyl bromide. (6 marks)

ii. Differentiate the reactivity of $LiAlH_4$ and $NaBH_4$ with one reaction as an example. (4marks)

(OR)

b) From the following given monomers, explain the synthesis of their polymers. (10marks)

i. Styrene and Butadiene ii. Ethylene glycol and terephthalic acid

iii.Ɛ amino caproic acid iv. Tetra fluoro ethylene

12. a) i. Write notes on Fibre reinforced composites with examples.

ii. Define Functionality and give its significance.

(OR)

b) Discuss in detail about the instrumentation and working of XPS with a neat diagram. (10marks)

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SRM

SET 1

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

CLA 3

Program: B.Tech
Course Code & Title: 21CYB101J / Chemistry
Year & Sem: I/II

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d) Mass of the atoms

10. Minimum interplanar spacing required for Bragg's diffraction is _____

a) $\lambda/4$ b) $\lambda/2$ c) $4\lambda$ d) $2\lambda$

**Part-B (2 x 10= 20 Marks)**

11. a) i.Explain the mechanism followed in the formation of anti-Markovnikov's product n-propyl bromide. (6 marks)

ii. Differentiate the reactivity of $LiAlH_4$ and $NaBH_4$ with one reaction as an example. (4marks)

(OR)

b) From the following given monomers, explain the synthesis of their polymers. (10marks)

i. Styrene and Butadiene ii. Ethylene glycol and terephthalic acid

iii.Ɛ amino caproic acid iv. Tetra fluoro ethylene

12. a) i. Write notes on Fibre reinforced composites with examples.

ii. Define Functionality and give its significance.

(OR)

b) Discuss in detail about the instrumentation and working of XPS with a neat diagram. (10marks)

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SRM

SET 1

SRM Institute of Science and Technology
College of Engineering and Technology
Department Of Chemistry
SRM Nagar, Kattankulathur – 603203

**CLA 3 Answer Key**

**Program: B.Tech**
**Course Code & Title: 21CYB101J/Chemistry**
**Year &Sem: I/II**

**Date: 04-05-2023**
**Time: 8.00-9.00am**
**Max. Marks: 30 Marks**

**Answer ALL the MCQs Part-A (10*1=10 Marks)**

1. b) Benzoyl peroxide

2. c) N-(4-hydroxy phenyl) acetamide

3. b) i and iv

4. d) Polyhexamethylene adipamate

5. b) Isoprene

6. b) Unsaturated compounds

7. d) Reinforcement

8. d) Automotive parts

9. c) Binding energy of the electron

10. b) $\lambda/2$

**Part-B ( 2x 10= 20 Marks)**

**11. a) i) Explain the mechanism followed in the formation of anti-Markovnikov's product n-propyl bromide.** **(6marks)**

Explanation of free radical mechanism of addition reaction with the Steps

**ii) Differentiate the reactivity of $LiAlH_4$ and $NaBH_4$ with one reaction as an example.** **(4marks)**

- $LiAlH_4$ is a strong reducing agent. It undergoes violent reaction with water.
- $NaBH_4$ is a mild reducing agent. It reacts slowly with water. While $NaBH_4$ reduces aldehydes and ketones rapidly, it reduces esters slowly. Hence, aldehyde and ketone can be reduced without simultaneous reduction of an ester group. But this is not possible in $LiAlH_4$. (2 Marks)

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- Any one reaction of differentiating the reactivity of $LiAlH_4$ and $NaBH_4$. (2 Marks)

(OR)

**b) How are the following polymers synthesized from the given monomers? (10marks)**

**i) Styrene and Butadiene** **ii) Ethylene glycol and terephthalic acid**

**iii) Ɛ amino caproic acid** **iv) Tetra fluoro ethylene**

i) Styrene and Butadiene------Synthesis of Buna-S rubber

Explanation---------- 1mark

Equation ---------- 1.5 marks

ii) Ethylene glycol and terephthalic acid----- Synthesis of Polyester

Explanation---------- 1mark

Equation ---------- 1.5 marks

iii) Ɛ amino caproic acid----------Synthesis of Nylon 6

Explanation---------- 1mark

Equation ---------- 1.5 marks

iv) Tetra fluoro ethylene----------Synthesis of Teflon

Explanation---------- 1mark

Equation ---------- 1.5 marks

**12. a) i) Write notes on Fibre reinforced composites with examples (5 marks)**

- Explanation of Any two types of fibre reinforced composites with examples— (5marks)

**ii. Define Functionality and give its significance.**

- Definition: Number of active sites in a monomer
  Example: Any one (Bifunctional or trifunctional monomer) (3marks)
- Significance of functionality Explanation on bifunctional, trifunctional and branched polymers (2 marks)

(OR)

**b) i) Discuss on the instrumentation and working of XPS.. (10marks)**

**Intrumentation**

Source, Sample holder, Energy analyser, Detector, UHV- (4 marks)

Diagram (2 marks)

Working (4 marks)

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SRM
INSTITUTE OF SCIENCE & TECHNOLOGY

SET 2

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

CLA 3

Program: B.Tech
Course Code & Title: 21CYB101J / Chemistry
Year &Sem: I / II

Date: 04-05-2023
Time: 8.00-9.00 am
Max. Marks: 30 Marks

**Answer ALL the MCQs Part-A (10 x 1=10 Marks)**

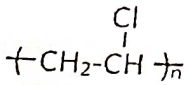
1. $S_N2$ reactions are

a) Usually occur through unimolecular b) Unimolecular c) Bimolecular d) both b and c

2. Aspirin is ------.

a) Acetylsalicylic acid b) Benzoyl salicylic acid c) Chlorobenzoic acid d) Anthranilic acid

3. The polymer shown is a _____.



a) Elastomer b) Fibre c) Thermoplastic d) Thermosetting plastic

4. Nylon-6 is made from

a) 1, 3-Butadiene b) chloroprene c) adipic acid d) caprolactam

5. The random orientation of the polymeric chain in a polymer is called

a) Isotactic b) Atactic c) Syndiotactic d) Elastomer

6. Weight average molecular weight _____ on the weight of molecules in a polymer.

a) Dependent b) Independent c) Partially dependent d) Neither dependent nor independent

7. ESCA can identify elements in the periodic table above which of the following element?

a) Carbon b) Boron c) Helium d) Potassium

8. Hooke's law essentially defines

a) Stress b) Strain c) Yield point d) Elastic limit

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9. Kevlar is a------ type of material.

a) Glass b) Thermoplastic c) Whisker d) Polymer

10. Which of the following technique is used to investigate the molecular structure through the growth of solid crystals?

a) Radiotherapy b) Fluoroscopy c) Computed Tomography d) X-ray crystallography

**Part-B (2 x 10= 20 Marks)**

11. a) i.Explain E2 mechanism by taking an example.

ii. Give the synthesis and uses of Paracetamol

(OR)

b) i. How polymers are classified based on origin and nomenclature? (6Marks)

ii. Suggest the products when 1, 3 – butadiene reacts with the following and provides suitable equations: (4Marks)

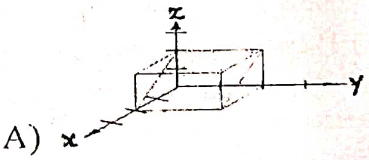
A) Acrylonitrile B) Styrene

12. a) i.At what glancing angle (θ), would first order reflection from (110) plane of KCl crystal be observed using X-ray of wavelength ( $\lambda$) = 145pm and dimension of unit cell is 315pm? (4Marks)

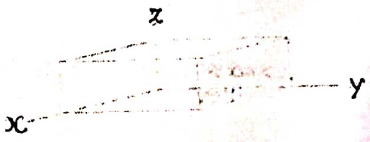
ii. How isotactic and atactic polymers differ? Give an example for each. (6Marks)

(OR)

b) i) Compute the Miller indices for the shaded planes given below: (6marks)

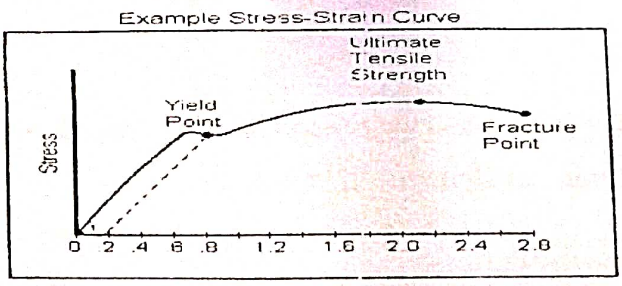
A) 


B)


C) 


ii. Explain the points given in stress-strain curve below:

(4Marks)

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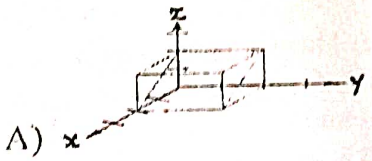
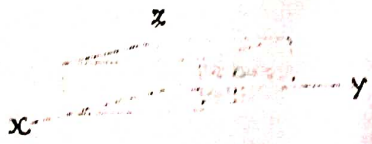
A) 


B)


C) 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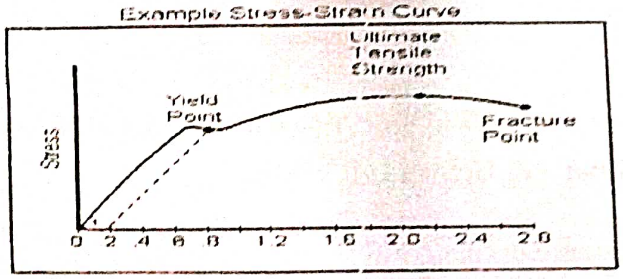


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SRM

INSTITUTE OF SCIENCE & TECHNOLOGY

SET 2

# DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

CLA 3

**Program: B.Tech**
**Course Code & Title: 21CYB101J / Chemistry**
**Year &Sem: I / II**

**Date: 04-05-2023**
**Time: 8.00-9.00 am**
**Max. Marks: 30 Marks**

**Answer ALL the MCQs Part-A (10 x 1=10 Marks)**

1. $S_N2$ reactions are

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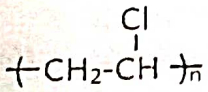

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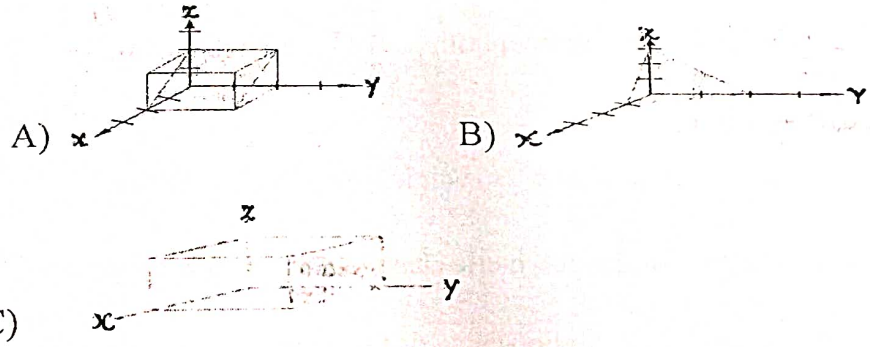
A) Acrylonitrile B) Styrene

12. a) i.At what glancing angle (θ), would first order reflection from (110) plane of KCl crystal be observed using X-ray of wavelength ( λ) = 145pm and dimension of unit cell is 315pm? (4Marks)

ii. How isotactic and atactic polymers differ? Give an example for each. (6Marks)

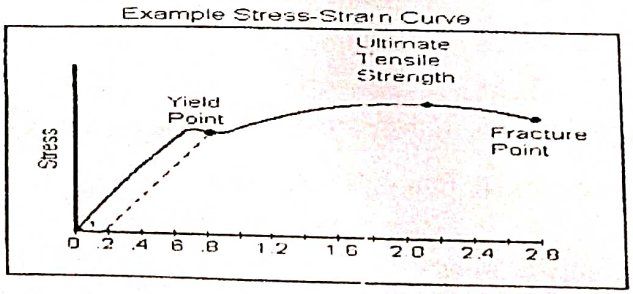
(OR)

b) i) Compute the Miller indices for the shaded planes given below: (6marks)




ii. Explain the points given in stress-strain curve below:

(4Marks)

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College of Engineering and Technology
Department of Chemistry
SRM Nagar, Kattankulathur – 603203

SET 2

CLA – III Answer Key

Program: B. Tech
Course Code & Title: 21CYB101J/Chemistry
Year &Sem: I/II

Date: 04-05-2023
Time: 8.00-9.00 am
Max. Marks: 30

**Part-A (10*1=10 Marks)**
**Answer ALL the MCQs**

1. c) Bimolecular
2. a) Acetylsalicylic acid
3. c) Thermoplastic
4. d) caprolactam
5. b) Atactic
6. a) Dependent
7. c) Helium
8. d) Elastic limit
9. d) Polymer
10. d) X-ray crystallography

**Part-B (2x 10= 20 Marks)**

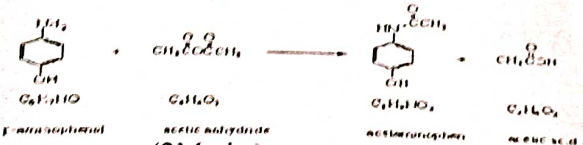
11. a) i. **Explain E2 mechanism by taking an example.**

- E2 mechanism is the bimolecular elimination mechanism, which the reaction rate depends on the concentration of both substrate and base.
- Explanation by taking an example.

(5Marks)

ii. **Explain the synthesis and uses of Paracetamol.**

- Synthesis - (3Marks)



- Any two uses – (2Marks)

**(OR)**

b) **i. How polymers are classified based on origin and nomenclature?** **(6Marks)**

Based on Origin (3 Marks)
Base on Nomenclature (3 Marks)

ii. **Suggest the products when 1,3-butadiene reacts with the following and provides suitable equations:** **(4Marks)**

A) With Acrylonitrile gives Buna-N rubber. Give equation. (2 Marks)

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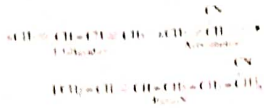
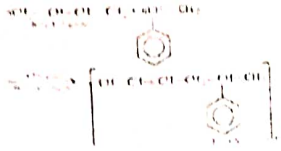


(2Marks)

B) With Styrene gives Buna-S rubber. Give equation.



12. a) i. **At what glancing angle (0), would first order reflection from (110) plane of KCl crystal be observed using X-ray of wavelength (λ) = 145pm and dimension of unit is 315pm?** **[4 Marks]**

nλ = 2dsin0; Where, n = 1, λ = 145, and d = 315

Substituting the values, we get, **0 = 13.3°**

**ii. How isotactic and atactic polymers differ? Give an example for each. (6 Marks)**

- Isotactic polymers all the substituents are located on the same side of the polymer chain.
- Atactic polymers the substituents are placed randomly along the chain. ( 4Marks)
- One Example for each. (2Marks)

(OR)

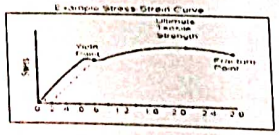
**b) i) Compute the Miller indices for the shaded planes given below: (6marks)**

A) B) C)

A) (303) B) (632) C) (010) (3x2=6 Marks)

Steps must be given.

ii. Explain the points given in stress-strain curve below: (4Marks)



- **Yield Point (1.5Marks)**

The yield point is defined as the point at which the material starts to deform plastically. After the yield point is passed, permanent plastic deformation occurs. There are two yield points (i) upper yield point (ii) lower yield point.

- **Ultimate Tensile Strength (1.5Marks)**

It is a point that represents the maximum stress that a material can endure before failure. Beyond this point, failure occurs.

- **Fracture or Breaking Point (1Marks)**

It is the point in the stress-strain curve at which the failure of the material takes place.

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Set-1

CLA – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 08.05.2023
Duration: 12.30 – 1.30 PM
Max. Marks: 30 Marks

**Part – A (10 x 1 = 10 Marks)**
**Answer ALL the Questions**

1. Markovnikov's law is applied in
   a) Addition of propylene with $Cl_2$
   b) Addition of propylene with HBr
   c) Addition of ethylene with $Br_2$
   d) Addition of ethylene with $H_2$

2. Which one of the following on reductio with Lithium Aluminium Hydride ($LiAlH_4$) yields a secondary amine?
   a) Methyl isocyanide
   b) Nitroethane
   c) Acetamide
   d) Methyl cyanide

3. Polydispersity index is defined as _____, where $M_w$ and $M_n$ are the weight average and number average molecular masses respectively.
   a) $M_w \times M_n$ b) $M_n/M_w$ c) $M_w - M_n$ d) $M_w/M_n$

4. What is Teflon?
   a) $(CF_2)_n$ b) $(C_2F)_n$ c) $(C_2F_4)_n$ d) $(C_4F_2)_n$

5. The characteristics of condensation polymerization are given below:
   I. only –C–C– linkages present in the polymer structure
   II. use of bifunctional or polyfunctional monomers
   III. elimination of a small by-product molecule

   Which of the following is true?
   a) I, II, III
   b) II and III
   c) I and III
   d) Only III

6. The non – metal used in the vulcanization of rubber is _____
   a) Phosphorous b) Graphite c) Silicon d) Sulphur

7. Which of the following statements is correct for ductile materials.
   a) Large deformation takes place between elastic limit and fracture point
   b) Have no proportional limit
   c) Break immediately after proportional limit
   d) Cannot be drawn into wires

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8. The continuous phase of a composite material is known as its_____
   a) dispersed phase b) surrounding phase
   c) matrix phase d) fiber phase

9. In the hemispherical electron analyser of XPS, _____ is detected and plotted as a function of energy.
   a) Mass b) Charge
   c) Number of electrons striking the detector d) Mass to charge ratio

10. If (3 2 6) are the Miller Indices of a plane, the intercepts made by the plane on the three crystallographic axes are
   a) (a, b, c) b) (2a, 3b, c) c) (a, 2b, 3c) d) (2a, b, 3c)

**Part – B (2 x 10 = 20 Marks)**

11. a. i. Explain the mechanism of Nucleophilic Substitution reaction, $S_N1$ with an example. (6 Marks)
   ii. Write the preparation and uses of the medicinal drug, Paracetamol. (4 Marks)

(OR)

b. i. Discuss the different types of tacticity shown by the polymers with example. (6 Marks)

ii. Write down the preparation, properties and uses of any one of the Polyamide fibres. (4 Marks)

12. a. i. Give the preparation, properties and uses of any one of the conducting polymers. (6 Marks)

ii. Arrive the equation $n\lambda = 2d\sin\theta$ used to predict the structure of crystals. (4 Marks)

(OR)

b. Elaborate the stress – strain relationship of solids with a neat plot. (10 Marks)

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Set-2

CLA – III

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 08.05.2023
Duration: 12.30 – 1.30 PM
Max. Marks: 30 Marks

**Part – A (10 x 1 = 10 Marks)**
**Answer ALL the Questions**

1. The rate of nucleophilic substitution reactions is higher in the presence of _____
   a) Electron withdrawing groups
   b) Electron releasing groups
   c) Both electron withdrawing and releasing groups
   d) Initiator

2. Oxidation of Ethene with cold alkaline $KMnO_4$ produces
   a) Formaldehyde
   b) $CO_2$ and $H_2O$
   c) Ethylene glycol
   d) Oxalic acid

3. Calculate the number average molecular mass of a polymer having four different monomers A, B, C and D present in equal number. The molecular masses of the monomers are 10000, 15000, 30000 and 50000 respectively.
   a) 10050
   b) 17350
   c) 26250
   d) 35475

4. A polymer with amide linkage is known as
   a) Terylene
   b) Teflon
   c) Bakelite
   d) Nylon-6,6

5. For which plastics can the process of heat softening, moulding and cooling to rigidness be repeated?
   a) thermosctting plastics
   b) thermoplastics
   c) bakelite
   d) Urea-formaldehyde

6. The monomer unit of natural rubber (polymer) is
   a) Isoprene
   b) Neoprene
   c) Chloroprene
   d) Butadiene

7. Which of the following statements is correct for brittle materials.
   a) It breaks soon after elastic limit is crossed
   b) It shows significant plastic deformation before breaking
   c) It is used to make wires
   d) Stress is never proportional to strain

8. Which of the following does not combine with fiber to give composites?
   a) Metals
   b) Ceramics
   c) Non-metals
   d) Polymers

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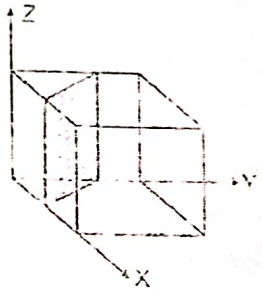
9. Which of the following is the most commonly used detector in X-ray Photoelectron Spectroscopy?

a) Electron multiplier
b) Dynodes
c) Photovoltaic cell
d) Photomultiplier

10. Miller Indices for the plane shown in the below figure is




a) (1 2 0)
b) (2 1 0)
c) (2 2 0)
d) (0 0 2)

## Part – B (2 x 10 = 20 Marks)

11. a. Write a short note on the following: (4+4+2 Marks)

i. Dieckmann condensation
ii. Oxidation of organic compounds by $KMnO_4$
iii. Ring opening reactions of cyclopropane with $H_2$ and HBr

**(OR)**

b. Discuss the n and p – doping mechanism in conducting polymers. (10 Marks)

12. a. i. How crosslinked polymers and linear polymers differ? Give an example for each. (4 Marks)

ii. Explain the principle of X-ray Photoelectron Spectroscopy and mention any two applications of it. (6 Marks)

**(OR)**

b. Brief about the particle reinforced and metal matrix composites with examples. (10 Marks)

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Set-2

## CLA – III
## ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 08.05.2023
Duration: 12.30 – 1.30 PM
Max. Marks: 30 Marks

### Part – A (10 x 1 = 10 Marks)
### Answer ALL the Questions

1. The rate of nucleophilic substitution reactions are higher in the presence of_____
   **a) Electron withdrawing groups**

2. Oxidation of Ethene with cold alkaline $KMnO_4$ produces
   **c) Ethylene glycol**

3. Calculate the number average molecular mass of a polymer having four different monomers A, B, C and D present in equal number. The molecular masses of the monomers are 10000, 15000, 30000 and 50000 respectively.
   **c) 26250**

4. A polymer with amide linkage is known as
   **d) Nylon-6,6**

5. For which plastics can the process of heat softening, moulding and cooling to rigidness be repeated?
   **b) thermoplastics**

6. The monomer unit of natural rubber (polymer) is
   **a) Isoprene**

7. Which of the following statements is correct for brittle materials.
   **a) It breaks soon after elastic limit is crossed**

8. Which of the following does not combine with fiber to give composites?
   **c) Non-metals**

9. Which of the following is the most commonly used detector in X-ray Photoelectron Spectroscopy?
   **a) Electron multiplier**

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10. Miller Indices for the plane shown in the below figure is

c) (2 2 0)

## Part – B (2 x 10 = 20 Marks)

11. a. Write a short note on the following: (4+4+2 Marks)

i. Dieckmann condensation
**Reaction with example – 4 Marks**

ii. Oxidation of organic compounds by $KMnO_4$
**Any 4 reactions – 4 Marks**

iii. Ring opening reactions of cyclopropane with $H_2$ and HBr – **Each 1 Mark**

**(OR)**

b. Discuss the n and p – doping mechanism in conducting polymers. (10 Marks)
**List of n & p – dopants – 2 Marks**
**Mechanism – Each 4 Marks**

12. a. i. How crosslinked polymers and linear polymers differ? Give an example for each. (4 Marks)

**Any 3 differences – 3 Marks; Example for each – 1 Mark**

ii. Explain the principle of X-ray Photoelectron Spectroscopy and mention any two applications of it. (6 Marks)
**Principle – 3 Marks; Diagram – 1 Mark; Any two applications – 2 Marks**

**(OR)**

b. Brief about the particle reinforced and metal matrix composites with examples. (10 Marks)

**Each Composite with description about Fillers, Matrix, examples, properties and applications – 5 Marks**

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# DEPARTMENT OF CHEMISTRY

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SET 1

CLA 3

Program: B.Tech
Course Code & Title: 21CYB101J / Chemistry
Year & Sem: I/II

Date: 04-05-2023
Time: 12.30-1.30pm
Max. Marks: 30

**Part-A (10x1=10 Marks)**

**Answer ALL the MCQs**

1. The reactivity order of alkyl halides in $S_N2$ mechanism is

a) $CH_3X > 1^0 > 2^0 > 3^0$ b) $CH_3X > 2^0 > 1^0 > 3^0$ c) $CH_3X > 3^0 > 1^0 > 2^0$ d) $CH_3X > 3^0 > 2^0 > 1^0$

2. The product of Dieckmann condensation reaction is

a) Cyclic alcohol b) β keto esters c) Cyclic ketone d) Alkane

3. Which of the following are the characteristics of thermosetting polymers?

i) Heavily branched cross linked polymers ii) Linear slightly branched long chain molecules

iii) Become infusible on moulding, so cannot be reused iv) Soften on heating and harden on cooling, can be reused

a) ii and iv b) i only c) iii only d) i and iii

4. The type of linkage present in poly urethane is

a) Amide linkage b) Glycosidic linkage c) Ester linkage d) Phospho diester linkage

5. Terylene is a

a) Poly ester b) Poly ethylene c) Poly propylene d) Poly amide

6. cis polyisoprene units present in natural rubber is an example of---- polymer.

a) Isotactic b) Syndiotactic c) Atactic d) both a and b

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7. Longitudinal strength of ...

a) Fibre strength b) Interface strength c) Elastic modulus d) Wear resistance

8. Which of the following does not combine with fibre to give composites?

a) Metals b) Ceramics c) Non-metals d) Polymers

9. The energy required to remove an electron from the highest occupied atomic orbital is

a) Kinetic energy b) Ionization energy c) Binding energy d) Vibrational energy

10. Calculate the Miller Indices of the plane, whose intercepts along the axes are (a,2b,3c)

a) (1 2 3) b) (3 2 1) c) (2 3 6) d) (6 3 2)

**Part-B (2x 10= 20 Marks)**

11. a) i) Discuss in detail about $S_N1$ mechanism in detail with an example. (6marks)

ii) Complete the following reactions (4marks)

1) $CH_2 = CH_2 \xrightarrow{cold.dil.KMnO_4/H_2O} ?$

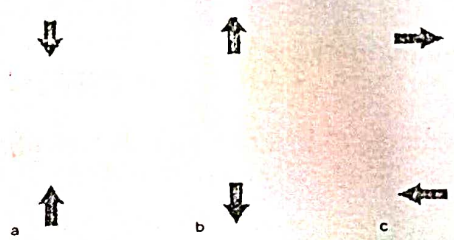
2) $CH_3\text{-}CH_2\text{-}C{\equiv}N \xrightarrow[LiAlH_4/H_2O]{} ?$

**(OR)**

b) Write the preparation, properties and uses of synthetic rubber and polystyrene. (10 marks)

12. a) i. Brief on extrinsically conducting polymers. (4 marks)

ii. From the figures given below, identify and define the different types of stress. (6Marks)




**(OR)**

b) i. Discuss on principle and applications of XPS. (6Marks)

ii) What is inter-planar spacing's in lattices? Give the expression taking an example. (4marks)

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SET 2

# DEPARTMENT OF CHEMISTRY

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CLA 3

Program: B.Tech
Course Code & Title: 21CYB101J / Chemistry
Year & Sem: I/II

Date: 04-05-2023
Time: 12.30-1.30pm
Max. Marks: 30

Part-A (10 x 1=10 Marks)

Answer ALL the MCQs

1. In $S_N1$ the first step involves the formation of_____
   a) Free radical b) Carbanion c) Final product d) Carbocation

2. Markovnikov's law is applied in addition of ----
   a) Propylene with $Cl_2$ b) Propylene with HBr c) Ethylene with $Br_2$ d) Ethylene with HCl

3. Heating of rubber with sulphur is known as:
   a) Galvanization b) Bessemerisation c) Vulcanization d) Sulphonation

4. The S-in Buna-S refers to
   a) Sulphur b) Styrene c) Sodium d) Salicylate

5. Which of the following is used for making rechargeable batteries?
   a) Polypyrrole b) Polyester c) Polyaniline d) Polyacrylonitrile

6. The non-stick layer of kitchenware contains:
   a) Teflon b) Acrilan c) Dacron d) Nylon

7. The property of a body by virtue of which it tends to regain its original size and shape when the applied force is removed is called
   a) Elasticity b) Plasticity c) Rigidity d) Compressibility

8. The continuous phase of a composite material is known as its ----------
   a) Dispersed phase b) Surrounding phase c) Matrix phase d) Fiber phase

9. In XPS process, the photon ejects which of the following?
   a) 1s electron b) 1p electron c) 2s electron d) 2p electron

10. If the angle of incidence is 30°, then the wavelength for first-order spectrum is equal to
    a) 2d b) d/2 c) d/3 d) d

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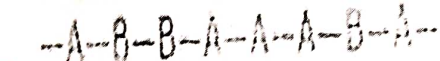
## Part – B (2 x 10 = 20 Marks)

11. a) i. What is the reaction of the following with Cyclopropane? (4 Marks)
A. Halogens B. HI C. sulphuric acid D. Hydrogen

ii. Explain Dieckmann condensation with an example. (6 Marks)

(OR)

b) i. Identify the type of polymer for the following and write a note on it. (4Marks)

A. B. 

ii. Discuss briefly on n and p doping in conducting polymers. (6Marks)

12. a) i. Write notes on metal matrix composites.

ii. Give the differences between Thermoplastic and Thermosets.

(OR)

b) Explain Bragg's law. How its principle is applied on studying diffraction of X- rays by atoms in a crystalline structure with a neat sketch.

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SET 1

CLA – III

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Semester: I Year & II Semester

Date: 08/05/2023
Duration: 8.00-9.00 am
Max. Marks: 30 Marks

**Part- A (10 x 1 = 10 Marks)**

**Answer ALL the Questions**

1. Identify the compound with the highest ring strain
   a) Cyclomethane
   b) Cyclopropane
   c) Cyclobutane
   d) Cyclopentane

2. Markovnikov's law is applied in
   (a) addition of propylene with $Cl_2$ (b) addition of propylene with HBr
   (b) addition of ethylene with $Br_2$ (d) addition of ethylene with HCl

3. Which of the following is true for the resultant polymer product formed, when molecules of phthalic acid react with molecules of glycerol?
   (a) branched polymer
   (b) cross-linked polymer
   (c) linear polymer
   (d) none of the mentioned

4. The polymer in which steric placements of the substituents are arranged in such a way to give alternate d and l configurations, is known as _____
   (a) isotactic polymer
   (b) atactic polymer
   (c) syndio-tactic polymer
   (d) none of the mentioned

5. The compound $[-CH_2-CH(C_6H_5)-]_n$ is a _____
   (a) homopolymer (b) co-polymer (c) condensation polymer (d) network polymer

6. Which of the following has the weakest intermolecular forces?
   (a) Buna-N (b) Nylon-6,6 (c) Polythene (d) Polystyrene

7. Which of the following wavelengths falls in an X-ray region
   (a) $10^{-4}$ Å (b) 1 Å (c) 1000 Å (d) 10000 Å

8. How is carbon fibre manufactured?



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(a) Heating (b) Thermal degradation of PAN
(b) Cooling of PAN (d) Thermal degradation of polyethylene

9. What term is used for the maximum stress at which material fails on a specified number of cycle?
(a) Fatigue strength (b) Fatigue life
(c) Ultimate tensile strength (d) Endurance limit

10. In the X-ray diffraction of a set of crystal planes having 'd' equal to 0.18 nm, the first-order reflection is found to be at an angle of 22°. The wavelength of X-rays is: (sin 22° = 0.208)
(a) 0.0749 nm (b) 0.0374 nm (c) 0.779 nm (d) 0.374 nm

**Part- B (2 x 10 = 20 Marks)**

11 a. Define oxidation and reduction reaction. Give examples of oxidation/reduction reaction using following reagents:
(A) $KMnO_4$ (B) $K_2Cr_2O_7$ (C) $NaBH_4$ (D) $LiAlH_4$ (10 Marks)

**OR**

b. i. How polymers are classified based on origin? Give examples. (4 Marks)

ii. What are conducting polymers? Explain n-doping and p-doping in case of conducting polymers with examples. (6 Marks)

12 a. i. Explain synthesis, properties and applications of:
(A) Polyacetylene (B) Teflon (6 Marks)

ii. What are particle reinforced composites? Give examples. (4 Marks)

**OR**

b. i. What is principle of X-ray photoelectron Spectroscopy (XPS)? Explain with neat diagram. (4 Marks)

ii. Explain stress-strain relationship plot for solids with suitable diagram. (6 Marks)

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SET 2

CLA – III

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 08/05/2023
Duration: 8.00-9.00 am
Max. Marks: 30 Marks

**Part- A (10 x 1 = 10 Marks)**

**Answer ALL the Questions**

1. Which of the following reacts by the E1 mechanism in ethanol most readily?
(a) $CH_3CH_2CH_2CH_2Br$ (b) $(CH_3)_2CHCH_2Br$ (c) $(CH_3)_3CBr$ (d) $CH_3CH_2CHBrCH_3$

2. Identify the **incorrect** statement regarding cycloalkanes.
(a) These have $sp^3$ hybridized carbons
(b) These have tetrahedral bond angles
(c) Stability of the cycloalkanes varies directly with their respective size
(d) These undergo nucleophilic substitution reactions

3. The characteristics of condensation polymerization are given below-

I. only –C–C– linkages present in the polymer structure
II. use of bifunctional or polyfunctional monomers
III. elimination of a small byproduct molecule

Which of the following is true?
(a) I, II, III (b) II and III (c) I and II (d) Only III

4. The strength of the polymer increases with _____ in molecular weight.
(a) Increase (b) Decrease (c) No change (d) Slightly decrease

5. Which of the following is used for making rechargeable batteries?
(a) Polypyrrole (b) Polyester (c) Polyaniline (d) Polyacrylonitrile

6. _____ undergoes permanent deformation on heating.
(a) Polythene (b) PVC (c) Teflon (d) Bakelite

7. Which of the following represents volumetric strain?
(a)-ΔV/V (b) ΔV/V ( c) Pressure/Volume (d) -P / (ΔV/V)

8. The kinetic energy of the photoelectron is dependent on _____ of the atom, which makes XPS useful to identify the oxide state.
(a) Mass (b) Charge (c) Chemical environment (d) Volume



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9. _____ is the measure of permanent deformation due to friction from a sharp object.
(a) Rebound hardness (b) Indentation hardness
(c) Toughness (d) Scratch hardness

10. The Miller indices of the plane, whose intercepts along the axes are (2a, b, 3/2c) are
(a) (3 6 4) (b) (6 3 4) (c) (6 3 2) (d) (3 6 2)

**Part- B (2 x 10 = 20 Marks)**

11 a. Explain $S_N1$ reaction mechanism with suitable example. How does the nature of the substrate affect the rate of the reaction? Comment on the stereochemistry of the reaction. (10 Marks)

**OR**

b. i. Explain synthesis, properties and applications of:
(A) PVC (B) Nylon 6, 6 (6 Marks)

ii. Compare addition polymerization with condensation polymerization with examples. (4 Marks)

12 a. i. How isotactic and syndiotactic polymers differ? Give examples. (4 Marks)

ii. Write a note on metal matrix composites. (6 Marks)

**OR**

b. i. Derive Bragg's law of X-ray diffraction. ( 6Marks)

ii. Explain the following terms: ( 4 Marks)
(A) Ultimate strength (B) Tensile strength

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